SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
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Section / Batch:	2025	**Date:**	16/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:	**100 Marks**	

SECTION A – APPLICATION BASED QUESTIONS

Q1. ***Automated Document Processing System***

A document processor handles one page at a time using the recurrence $T(n) = T(n-1) + 1$. Recall the meaning of a recurrence relation, explain how it expands step-by-step, and solve it using substitution method to determine the final time complexity. Also determine the space complexity of the recursive process.

Q2. ***Satellite Data Transmission System***

A satellite system divides transmitted data using the recurrence $T(n) = 2T(n/2) + n$. Apply the Master Theorem to determine the time complexity. Identify parameters and also determine the space complexity of the recursion.

Code of Conduct

I Beffy A Shanika(192511387) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: _________________	Signature: _________________	Signature: _________________
Date: _____________________	Date: _____________________	Date: _____________________

Q1.Introduction

An Automated Document Processing System processes documents by handling one page at a time. Suppose a document contains n pages. The system first processes one page and then recursively processes the remaining (n−1) pages until no pages are left.

The time required to process the document is represented by the recurrence relation:

$$T(n) = T(n - 1) + 1$$

where:

- T(n) = Time taken to process n pages
- T(n−1) = Time taken to process the remaining (n−1) pages
- +1 = Constant time required to process one page

This recurrence shows that the algorithm performs the same operation repeatedly for each page.

Meaning of a Recurrence Relation

A recurrence relation is a mathematical equation that defines the running time of an algorithm in terms of its running time on a smaller input.

Instead of directly calculating the total execution time, it breaks the problem into smaller subproblems until a base case is reached.

For this problem,

$$T(n) = T(n - 1) + 1$$

means:

- The system processes one page.
- Then it recursively processes the remaining pages.
- This continues until only one page is left.

Expansion of the Recurrence Relation

Expand the recurrence repeatedly.

Step 1

$$T(n) = T(n - 1) + 1$$

Step 2

Substitute

$$T(n - 1) = T(n - 2) + 1$$

Therefore,

$$T(n) = T(n - 2) + 1 + 1T(n) = T(n - 2) + 2$$

Step 3

Substitute

$$T(n - 2) = T(n - 3) + 1$$

Then,

$$T(n) = T(n - 3) + 3$$

Step 4

Substitute again,

$$T(n) = T(n - 4) + 4$$

Continuing this process,

$$T(n) = T(n - k) + k$$

When the recursion reaches the base case,

$$n - k = 1$$

Therefore,

$$k = n - 1$$

Substitute this value,

$$T(n) = T(1) + (n - 1)$$

Solving Using the Substitution Method

Assume the base case:

$$T(1) = 1$$

Substitute into the equation,

$$T(n) = 1 + (n - 1)T(n) = n$$

Hence,

$$\boxed{T(n) = n}$$

Time Complexity

The solution obtained is

$$T(n) = n$$

Ignoring constants and lower-order terms,

$$\boxed{T(n) = O(n)}$$

Interpretation

The processor handles one page at a time.

If there are:

- 10 pages → 10 processing steps
- 100 pages → 100 processing steps
- 1000 pages → 1000 processing steps

Thus, the running time increases linearly with the number of pages.

Time Complexity Table

Case	Time Complexity
Best Case	**O(n)**
Average Case	**O(n)**
Worst Case	**O(n)**

Since every page must be processed, all three cases require linear time.

Space Complexity

The algorithm is recursive.

Each recursive call is stored in the call stack until the base case is reached.

For n pages, there will be n recursive calls.

Example:

```
T(5)
|
├──T(4)
|   ├──T(3)
|   |   ├──T(2)
|   |   |   └──T(1)
```

The recursion stack contains 5 function calls before returning.

Therefore,

$$\boxed{\text{Space Complexity} = O(n)}$$

Final Result

- **Recurrence Relation**

$$T(n) = T(n - 1) + 1$$

- **Expanded Form**

$$T(n) = T(n - 2) + 2T(n) = T(n - 3) + 3T(n) = T(n - 4) + 4T(n) = T(1) + (n - 1)$$

- **Substitution Method**

$$T(n) = 1 + (n - 1)\boxed{T(n) = n}$$

- **Time Complexity**

$$\boxed{O(n)}$$

- **Space Complexity**

$$\boxed{O(n)}$$

Q2.Introduction

In a Satellite Data Transmission System, large amounts of data such as satellite images, weather information, GPS signals, and communication packets are transmitted from satellites to Earth.

To improve transmission efficiency, the system uses the Divide and Conquer technique. The data is divided into two equal parts, each part is transmitted or processed separately, and finally the results are combined.

The recurrence relation representing this process is:

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

where:

- T(n) = Time required to process n units of data
- 2T(n/2) = The data is divided into two equal halves, and each half is processed recursively.
- +n = Time required to combine or merge the processed data.

Meaning of the Recurrence Relation

The recurrence

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

indicates that:

- The satellite divides the data into two equal parts.
- Each part is processed independently.
- After processing, both parts are merged together.
- This process continues recursively until the data size becomes very small.

This is similar to the Merge Sort divide-and-conquer strategy.

Applying the Master Theorem

The standard form of the Master Theorem is:

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

Compare with the given recurrence:

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

Identify the Parameters

Parameter	Value	Meaning
A	**2**	Number of recursive subproblems
b	**2**	Each subproblem size is reduced by half
f(n)	**n**	Time required to combine the results

Calculate $n^{\log_b a}$

Here,

$$a = 2, b = 2$$

Therefore,

$$\log_b a = \log_2 2 = 1$$

Hence,

$$n^{\log_b a} = n^1 = n$$

Since

$$f(n) = n$$

and

$$n = n^{\log_b a}$$

we have

$$f(n) = \Theta\left(n^{\log_b a}\right)$$

Applying Master Theorem (Case 2)

According to the Master Theorem:

If

$$f(n) = \Theta\left(n^{\log_b a} \log^k n\right)$$

where $k = 0$,

then

$$T(n) = \Theta\left(n^{\log_b a} \log^{k+1} n\right)$$

Substituting the values,

$$T(n) = \Theta(n \log n)$$

Therefore,

$$\boxed{T(n) = O(n\log n)}$$

Time Complexity

The algorithm divides the satellite data into two equal parts repeatedly and combines the processed data at each level.

Hence,

$$\boxed{\text{Time Complexity} = O(n\log n)}$$

Interpretation

Suppose the satellite has:

- 1024 data packets

The packets are divided as follows:

Level 0 : 1024

Level 1 : 512 + 512

Level 2 : 256 + 256 + 256 + 256

Level 3 : 128 + 128 + ...

The division continues until only one packet remains.

There are approximately $\log_2 n$ levels, and each level processes n packets.

Therefore,

$$n + n + n + \cdots (\log n \text{ levels}) = n\log n$$

Space Complexity

The recursive calls create a recursion stack.

The depth of recursion is

$$\log_2 n$$

Only one recursive path is active at a time, so the recursion stack stores approximately log n function calls.

Hence,

$$\boxed{\text{Space Complexity} = O(\log n)}$$

Final Result

Recurrence Relation

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

Master Theorem Parameters

- $a = 2$

- $b = 2$
- $f(n) = n$

Calculation

$$n^{\log_2 2} = n$$

Since

$$f(n) = \Theta(n)$$

Master Theorem Case 2 applies.

Time Complexity

$$\boxed{O(n\log n)}$$

Space Complexity

$$\boxed{O(\log n)}$$